TEMPERATE CONTINENTAL STEPPE CLIMATE

TEMPERATE GRASSLAND WORLD MAP → CONTINENTALITY MECHANISM → RAINFALL-MOISTURE PROFILE → HEMISPHERE ASYMMETRY → CHERNOZEM-SOIL CHAIN → GRANARY CONVERSION TIMELINE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

★ PRIMARY CORE FLOW · ● SUBORDINATE EXTRA · ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g5 and all prior artifacts unchanged

00

STAGE 00

TEMPERATE GRASSLAND WORLD MAP

TEMPERATE GRASSLAND WORLD MAP · UKRAINE-KAZAKHSTAN BELT, CONTINENTAL INTERIOR · NORTH AMERICA · GREAT PLAINS, USA AND CANADA · SOUTH AMERICA · ARGENTINA AND URUGUAY, MARITIME FRINGE · SOUTH AFRICA · ELEVATED INTERIOR PLATEAU

EURASIA → steppes: Ukraine-Kazakhstan belt, continental interior

NORTH AMERICA → prairies: Great Plains, USA and Canada → SOUTH AMERICA → pampas: Argentina and Uruguay, maritime fringe → SOUTH AFRICA → veld: elevated interior plateau

CORE CLAIM
- EURASIA → steppes: Ukraine-Kazakhstan belt, continental interior
- NORTH AMERICA → prairies: Great Plains, USA and Canada
- SOUTH AMERICA → pampas: Argentina and Uruguay, maritime fringe

MECHANISM
- SOUTH AFRICA → veld: elevated interior plateau
- AUSTRALIA → downs: eastern interior grasslands
- MUST REMEMBER: Temperate continental steppe Bs climates occupy interiors and rain shadows between deserts and

CONSEQUENCE / LIMIT
- humid forests, with low variable precipitation, large annual range, short grasses and chernozem-linked grain/pastoral economies.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Temperate continental steppe Bs climates occupy interiors and rain shadows between deserts and humid forests, with low variable precipitation, large annual range, short grasses and chernozem-linked grain/pastoral economies.

01

STAGE 01

CONTINENTALITY MECHANISM

CONTINENTALITY MECHANISM · DISTANCE FROM SEA · WEAK MARITIME MODERATION · STRONG HEATING OF CONTINENTAL MASS · STRONG RADIATIVE COOLING · LARGE ANNUAL TEMPERATURE RANGE, ESPECIALLY IN EURASIA

DISTANCE FROM SEA → weak maritime moderation → SUMMER → strong heating of continental mass → hot → WINTER → strong radiative cooling → cold → RESULT → large annual temperature range, especially in Eurasia

CORE CLAIM
- DISTANCE FROM SEA → weak maritime moderation
- SUMMER → strong heating of continental mass → hot

MECHANISM
- WINTER → strong radiative cooling → cold
- RESULT → large annual temperature range, especially in Eurasia

CONSEQUENCE / LIMIT
- DISTANCE FROM SEA → weak maritime moderation
- SUMMER → strong heating of continental mass → hot

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → large annual temperature range, especially in Eurasia.

02

STAGE 02

RAINFALL-MOISTURE PROFILE

RAINFALL-MOISTURE PROFILE · TOTAL ANNUAL · ABOUT 25-75 CM, LOW TO MODERATE · SEASONAL MAXIMUM · LATE SPRING AND EARLY SUMMER · DRY MARGIN · SHORTER GRASS, STEPPE-DESERT TRANSITION · WET MARGIN

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
TOTAL ANNUAL → about 25-75 cm, low to moderate	DRY MARGIN → shorter grass, steppe-desert transition	TOTAL ANNUAL → about 25-75 cm, low to moderate
SEASONAL MAXIMUM → late spring and early summer	WET MARGIN → taller grass, steppe-forest boundary	SEASONAL MAXIMUM → late spring and early summer

CORE CLAIM
- TOTAL ANNUAL → about 25-75 cm, low to moderate
- SEASONAL MAXIMUM → late spring and early summer

MECHANISM
- DRY MARGIN → shorter grass, steppe-desert transition
- WET MARGIN → taller grass, steppe-forest boundary

CONSEQUENCE / LIMIT
- TOTAL ANNUAL → about 25-75 cm, low to moderate
- SEASONAL MAXIMUM → late spring and early summer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEASONAL MAXIMUM → late spring and early summer.

03

STAGE 03

HEMISPHERE ASYMMETRY

HEMISPHERE ASYMMETRY · STEPPES AND PRAIRIES · LARGE CONTINENTAL MASS, EXTREME RANGE · PAMPAS, VELD, DOWNS · NARROWER LANDMASS, MARITIME MODERATION · CONTINENTAL MASS DETERMINES WINTER SEVERITY · EXAM TRAP · DO NOT TREAT ALL FIVE AS CLIMATICALLY IDENTICAL

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
NORTHERN → steppes and prairies: large continental mass, extreme range	CAUSE → continental mass determines winter severity	NORTHERN → steppes and prairies: large continental mass, extreme range
SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation	EXAM TRAP → do not treat all five as climatically identical	SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

CORE CLAIM
- NORTHERN → steppes and prairies: large continental mass, extreme range
- SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

MECHANISM
- CAUSE → continental mass determines winter severity
- EXAM TRAP → do not treat all five as climatically identical

CONSEQUENCE / LIMIT
- NORTHERN → steppes and prairies: large continental mass, extreme range
- SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXAM TRAP → do not treat all five as climatically identical.

04

STAGE 04

CHERNOZEM-SOIL CHAIN

CHERNOZEM-SOIL CHAIN · GRASSLAND COVER · DEEP ROOT NETWORK DECAYS IN PLACE · LOW LEACHING · BASE-RICH CONDITIONS PRESERVED · HUMUS ACCUMULATES · DARK, DEEP, FERTILE SOIL PROFILE · CHERNOZEM OR BLACK EARTH, AMONG WORLD'S MOST FERTILE

GRASSLAND COVER → deep root network decays in place → LOW LEACHING → base-rich conditions preserved → HUMUS ACCUMULATES → dark, deep, fertile soil profile → RESULT → chernozem or black earth, among world's most fertile

CORE CLAIM
- GRASSLAND COVER → deep root network decays in place
- LOW LEACHING → base-rich conditions preserved

MECHANISM
- HUMUS ACCUMULATES → dark, deep, fertile soil profile
- RESULT → chernozem or black earth, among world's most fertile

CONSEQUENCE / LIMIT
- GRASSLAND COVER → deep root network decays in place
- LOW LEACHING → base-rich conditions preserved

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → chernozem or black earth, among world's most fertile.

05

STAGE 05

GRANARY CONVERSION TIMELINE

GRANARY CONVERSION TIMELINE · NATURAL STATE · PASTORAL GRASSLAND, LOW-DENSITY USE · RAIL + OCEAN FREIGHT · ACCESS TO DISTANT INDUSTRIAL MARKETS · MECHANISATION + HYV · HIGH OUTPUT PER WORKER ON LEVEL PLAINS · WORLD'S WHEAT GRANARIES

NATURAL STATE → pastoral grassland, low-density use → RAIL + OCEAN FREIGHT → access to distant industrial markets → MECHANISATION + HYV → high output per worker on level plains → RESULT → world's wheat granaries: Ukraine, prairies, pampas

CORE CLAIM
- NATURAL STATE → pastoral grassland, low-density use
- RAIL + OCEAN FREIGHT → access to distant industrial markets

MECHANISM
- MECHANISATION + HYV → high output per worker on level plains
- RESULT → world's wheat granaries: Ukraine, prairies, pampas

CONSEQUENCE / LIMIT
- NATURAL STATE → pastoral grassland, low-density use
- RAIL + OCEAN FREIGHT → access to distant industrial markets

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → world's wheat granaries: Ukraine, prairies, pampas.

06

STAGE 06

ANTI-DETERMINISM FIREWALL

ANTI-DETERMINISM FIREWALL · PHYSICAL BASE · FERTILE SOIL + LEVEL RELIEF + SUITABLE CLIMATE · CONVERSION AGENTS · RAIL, FREIGHT, MECHANISATION, MARKET DEMAND · WITHOUT ACCESS · SAME SOIL, NO GRANARY FOR CENTURIES · ENDOWMENT IS NECESSARY, ACCESS IS SUFFICIENT

PHYSICAL BASE → fertile soil + level relief + suitable climate

CONVERSION AGENTS → rail, freight, mechanisation, market demand → WITHOUT ACCESS → same soil, no granary for centuries → CONCLUSION → endowment is necessary, access is sufficient

CORE CLAIM
- PHYSICAL BASE → fertile soil + level relief + suitable climate
- CONVERSION AGENTS → rail, freight, mechanisation, market demand

MECHANISM
- WITHOUT ACCESS → same soil, no granary for centuries
- CONCLUSION → endowment is necessary, access is sufficient

CONSEQUENCE / LIMIT
- PHYSICAL BASE → fertile soil + level relief + suitable climate
- CONVERSION AGENTS → rail, freight, mechanisation, market demand

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONCLUSION → endowment is necessary, access is sufficient.

07

STAGE 07

ECOLOGICAL-COST CHAIN

ECOLOGICAL-COST CHAIN · SOD REMOVAL · BARE FINE SOIL EXPOSED TO WIND · DROUGHT YEAR · VEGETATION FAILS, DEFLATION PEAKS · ORGANIC DECLINE · REDUCED HUMUS, COMPACTION RISK · AQUIFER DRAWDOWN

SOD REMOVAL → bare fine soil exposed to wind → DROUGHT YEAR → vegetation fails, deflation peaks → ORGANIC DECLINE → reduced humus, compaction risk → AQUIFER DRAWDOWN → irrigation in semi-arid belt depletes groundwater

CORE CLAIM
- SOD REMOVAL → bare fine soil exposed to wind
- DROUGHT YEAR → vegetation fails, deflation peaks

MECHANISM
- ORGANIC DECLINE → reduced humus, compaction risk
- AQUIFER DRAWDOWN → irrigation in semi-arid belt depletes groundwater

CONSEQUENCE / LIMIT
- SOD REMOVAL → bare fine soil exposed to wind
- DROUGHT YEAR → vegetation fails, deflation peaks

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AQUIFER DRAWDOWN → irrigation in semi-arid belt depletes groundwater.

08

STAGE 08

INDIA WHEAT GRANARY MAP

INDIA WHEAT GRANARY MAP · PUNJAB-HARYANA-WESTERN UP · INDO-GANGETIC ALLUVIAL PLAIN · ALLUVIAL, NOT CHERNOZEM (FUNDAMENTAL DIFFERENCE) · SOWN OCT-DEC, HARVESTED MAR-APR · HYV SEEDS + IRRIGATION + FERTILISER + MSP PROCUREMENT

PUNJAB-HARYANA-WESTERN UP → Indo-Gangetic alluvial plain

SOIL → alluvial, not chernozem (fundamental difference) → SEASON → Rabi: sown Oct-Dec, harvested Mar-Apr → ENABLERS → HYV seeds + irrigation + fertiliser + MSP procurement

CORE CLAIM
- PUNJAB-HARYANA-WESTERN UP → Indo-Gangetic alluvial plain
- SOIL → alluvial, not chernozem (fundamental difference)

MECHANISM
- SEASON → Rabi: sown Oct-Dec, harvested Mar-Apr
- ENABLERS → HYV seeds + irrigation + fertiliser + MSP procurement

CONSEQUENCE / LIMIT
- PUNJAB-HARYANA-WESTERN UP → Indo-Gangetic alluvial plain
- SOIL → alluvial, not chernozem (fundamental difference)

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ENABLERS → HYV seeds + irrigation + fertiliser + MSP procurement.

09

STAGE 09

GREEN REVOLUTION PACKAGE

GREEN REVOLUTION PACKAGE · HYV SEMI-DWARF WHEAT · RAISED YIELD POTENTIAL · CANAL + TUBE-WELL IRRIGATION · REDUCED RAINFALL DEPENDENCE · CHEMICAL FERTILISERS · SUPPORTED INTENSIVE CULTIVATION · ASSURED FLOOR PRICE AND CENTRAL POOL STOCKS

HYV SEMI-DWARF WHEAT → raised yield potential → CANAL + TUBE-WELL IRRIGATION → reduced rainfall dependence → CHEMICAL FERTILISERS → supported intensive cultivation → MSP/PROCUREMENT → assured floor price and central pool stocks → CLOSE DISTINCTION: Steppe differs from tropical savanna and humid prairie, chernozem is not universal across

CORE CLAIM
- HYV SEMI-DWARF WHEAT → raised yield potential
- CANAL + TUBE-WELL IRRIGATION → reduced rainfall dependence

MECHANISM
- CHEMICAL FERTILISERS → supported intensive cultivation
- MSP/PROCUREMENT → assured floor price and central pool stocks

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Steppe differs from tropical savanna and humid prairie, chernozem is not universal across all
- steppes, and continentality concerns land-dominated thermal range rather than simply distance from a national coastline.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Steppe differs from tropical savanna and humid prairie, chernozem is not universal across all steppes, and continentality concerns land-dominated thermal range rather than simply distance from a national coastline.

10

STAGE 10

SAVANNA-STEPPE COMPARISON

SAVANNA-STEPPE COMPARISON · TROPICAL SAVANNA VERSUS TEMPERATE STEPPE/PRAIRIE · WET-AND-DRY SEASONAL VERSUS LOW-TOTAL SPRING MAXIMUM · LEACHED LATERISING VERSUS BASE-RICH CHERNOZEM/PRAIRIE · PASTORAL-SUBSISTENCE VERSUS MECHANISED-COMMERCIAL GRAIN · EVIDENCE LIMIT · GRAIN PRODUCTIVITY DEPENDS ON SOIL, SNOWMELT, TECHNOLOGY AND MARKETS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
LATITUDE → tropical savanna versus temperate steppe/prairie	SOIL → leached laterising versus base-rich chernozem/prairie	EVIDENCE LIMIT: Grain productivity depends on soil, snowmelt, technology and markets as well as climate
RAINFALL → wet-and-dry seasonal versus low-total spring maximum	ECONOMY → pastoral-subsistence versus mechanised-commercial grain	India's wheat belt is a functional comparison with monsoonal irrigation and subtropical seasonality, not a climatic identity.

CORE CLAIM
- LATITUDE → tropical savanna versus temperate steppe/prairie
- RAINFALL → wet-and-dry seasonal versus low-total spring maximum

MECHANISM
- SOIL → leached laterising versus base-rich chernozem/prairie
- ECONOMY → pastoral-subsistence versus mechanised-commercial grain

CONSEQUENCE / LIMIT
- EVIDENCE LIMIT: Grain productivity depends on soil, snowmelt, technology and markets as well as climate
- India's wheat belt is a functional comparison with monsoonal irrigation and subtropical seasonality, not a climatic identity.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Grain productivity depends on soil, snowmelt, technology and markets as well as climate; India's wheat belt is a functional comparison with monsoonal irrigation and subtropical seasonality, not a climatic identity.

11

STAGE 11

STEPPE ANSWER SPINE

STEPPE ANSWER SPINE · CONTINENTAL INTERIOR GRASSLAND WITH LARGE ANNUAL RANGE · FIVE REGIONAL NAMES ON A WORLD MAP · MOISTURE DEFICIT, CHERNOZEM, GRANARY CONVERSION · ACCESS CONVERTED ENDOWMENT INTO RESOURCE

DEFINE → continental interior grassland with large annual range → LOCATE → five regional names on a world map → EXPLAIN → moisture deficit, chernozem, granary conversion → QUALIFY → anti-determinism: access converted endowment into resource

CORE CLAIM
- DEFINE → continental interior grassland with large annual range
- LOCATE → five regional names on a world map

MECHANISM
- EXPLAIN → moisture deficit, chernozem, granary conversion
- QUALIFY → anti-determinism: access converted endowment into resource

CONSEQUENCE / LIMIT
- DEFINE → continental interior grassland with large annual range
- LOCATE → five regional names on a world map

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → anti-determinism: access converted endowment into resource.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · INDIA'S WHEAT GRANARY · PUNJAB, HARYANA, WESTERN UP · INDO-GANGETIC PLAIN · PUNJAB IS DESCRIBED AS A MAJOR FOOD BASKET/GRANARY IN · PUNJAB'S HIGH CULTIVATED SHARE AND CROPPING INTENSITY ARE SOURCE-ERA · HARYANA'S WHEAT BELT SPANS IRRIGATED EASTERN, CENTRAL AND WESTERN · WHEAT IS A RABI CROP

OPTIONAL SOURCE DEPTH
- India's wheat granary = Punjab, Haryana, western UP / Indo-Gangetic plain.
- Punjab is described as a major food basket/granary in the source text.
- Punjab's high cultivated share and cropping intensity are source-era indicators, not current statistics.
- Haryana's wheat belt spans irrigated eastern, central and western plains; use current district maps.

ADVANCED DEBATE
- Wheat is a Rabi crop: sown Oct–Dec, harvested Mar–Apr in source notes.
- Green Revolution package: HYV + irrigation + fertiliser + MSP/procurement.
- Soil is alluvial , not steppe chernozem.
- The source-era text records a substantial marketed surplus; current share requires a dated source.

LEGACY / NUANCE
- Surplus states: Punjab, Haryana, UP, Rajasthan, MP.
- Deficit destinations include Maharashtra, West Bengal, Bihar and Delhi.
- NOT: India's wheat belt sits on chernozem like the steppe → It sits on Indo-Gangetic alluvial soil.
- NOT: Wheat is a Kharif monsoon crop → Wheat is a Rabi winter crop.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels